Es. 1)

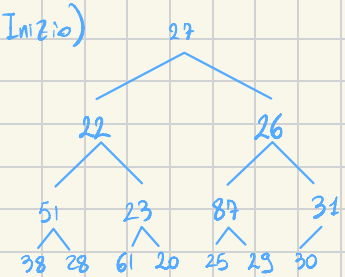
27 22 26 51 23 87 31 38 28 61 20 25 29 30

Trasformare in heap, con val minimo in radice. Eseguire poi i primi 2 passi di heap sort

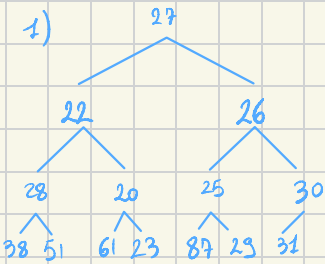
SVOLGIMENTO:

27 | 22 26 | 51 23 87 31 | 38 28 61 20 25 29 30

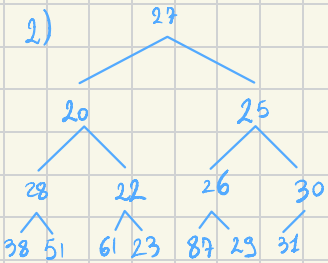
Inizio)



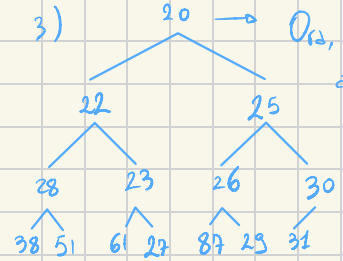
1)



2)

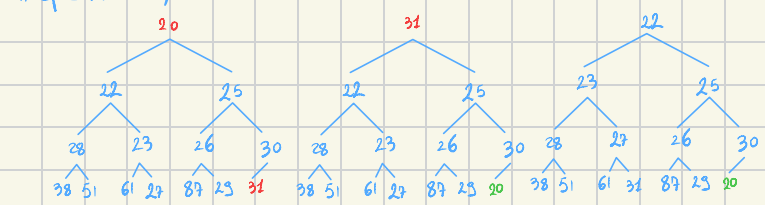


3)

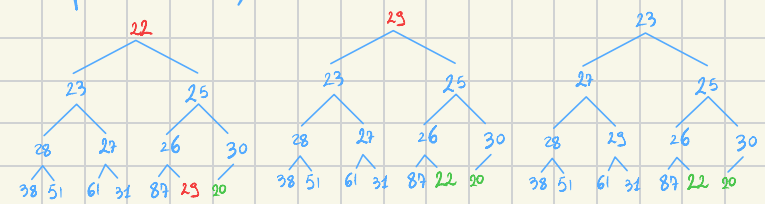


→ Ora, nella radice dello heap abbiamo il valore minimo.

Heap sort 1)



Heap Sort 2)

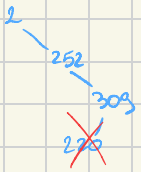


Es. 2) BST con num tra 0 e 400, cercare chiave 255. Quale corretta?

a) 2 252 309 220 330 264 255
b) 2 252 209 220 240 250 255
c) 1 400 202 300 270 250 255

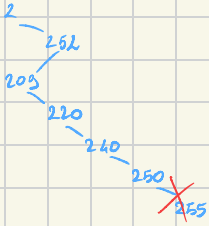
SVOLGIMENTO

a) 2 252 309 220 330 264 255



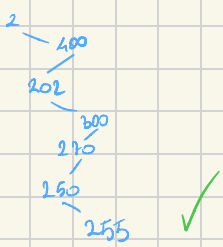
Errore: 220 < 309 quindi ramo sx di 309, ma siamo già nel ramo dx di 252, ma 220 < 252!

b) 2 252 209 220 240 250 255



Errore: 255 non può trovarsi nel ramo sinistro di 250, ovvero tra gli elementi minori di 252.

c) 1 400 202 300 270 250 255



Es. 3) Tabella hash. Open addressing con linear probing. $\alpha < 0.4$

Chiavi: 211 26 73 46 154 17 43 230 16

SVOLGIMENTO:

1) Dimensione tabella? Poiché $\alpha = \frac{N}{M} \Rightarrow M = \frac{N}{\alpha} = \frac{9}{0.4} = 22.5$ → Deve essere num primo → M = 23

2) Quale funzione? Escludiamo le opzioni a) e d) poiché quando abbiamo h_1 e h_2 stiamo usando double hashing, mentre qui si richiede linear probing con $M = 23$, quindi opzione b).

3) SVOLGIMENTO

0	46 (230 hit!)	5		10	99	15		20	43
1	230	6		11		16	154 (16 hit!)	21	
2		7		12		17	17 (16 hit!)	22	
3	26	8		13		18	16		
4	211	9		14		19			

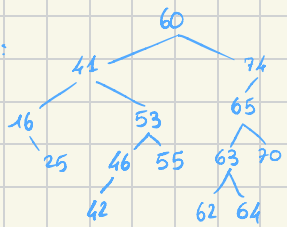
$h(K) = (K \cdot i) \,\%\, M$

Risposte alle domande:

1) Collisioni 16: cella 1 = 16, cella 2 = 17

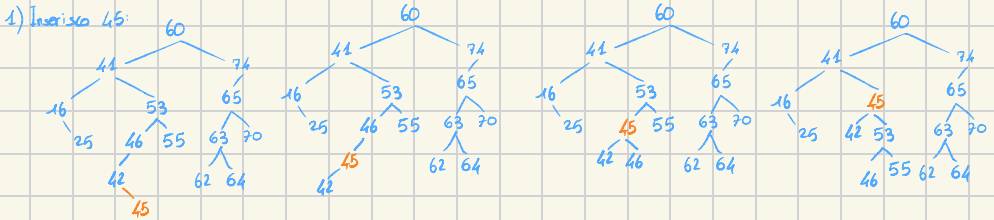
2) Posizione chiave 230? cella 1.

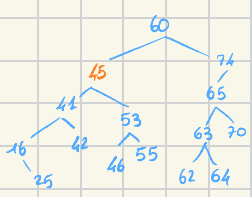
Es. 4) Inserire in radice 45 poi cancellare 60. BST:

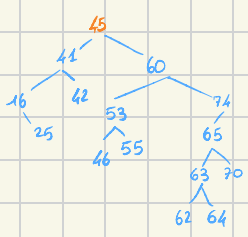


SVOLGIMENTO:

1) Inserisco 45:



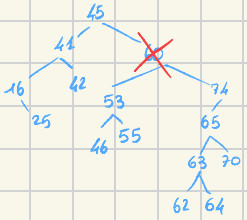


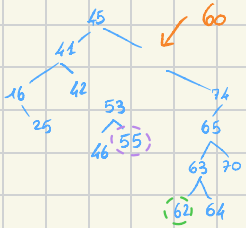


RISPOSTE ALLE DOMANDE:

- figlio SX: 41
- figlio DX: 60

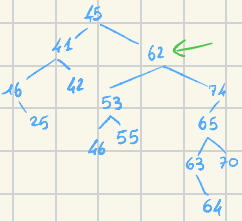
2) Cancellazione 60:





60 va sostituito con una delle due opzioni:

1. Il suo successore più stretto
2. Il suo antenato più stretto

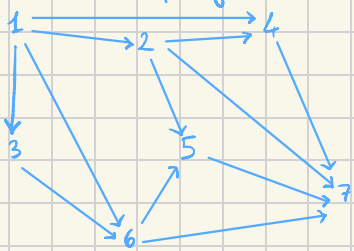


← Successore più stretto

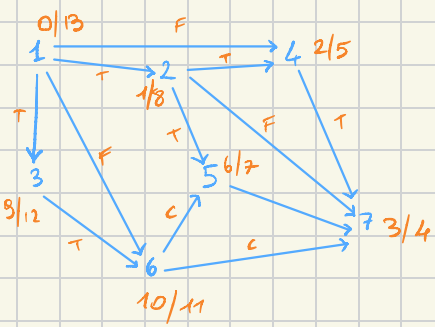
RISPOSTA ALLA DOMANDA:

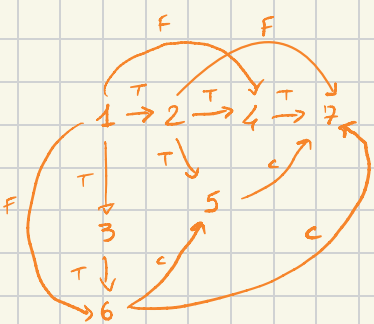
- figlio sx 65: 63
- figlio dx 65: 70

Es. 5) Ordinare topologicamente DAG. 1 come partenza.



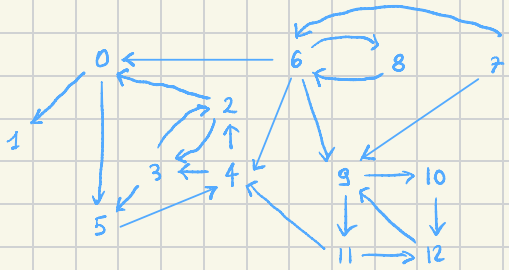
SVOLGIMENTO:





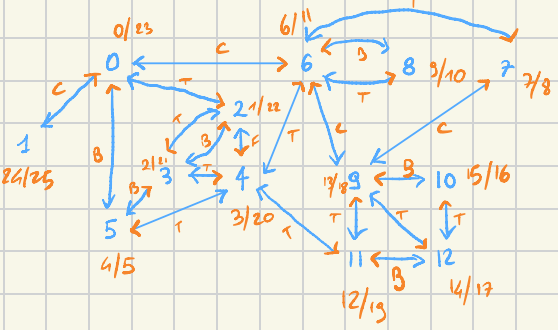
Ordinamento topologico (per tempo di terminazione decrescente): 1, 3, 6, 2, 5, 4, 7

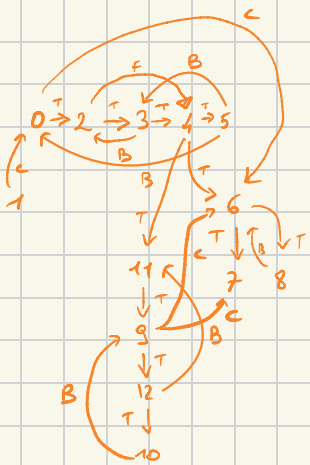
Es 6) Algoritmo Kosaraju per SCC, partendo da 0.



SVOLGIMENTO:

1) DFS su G^T

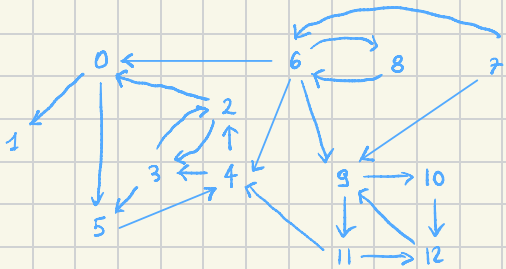




Ordino per t_{fine} decrescente:

1, 0, 2, 3, 4, 11, 9, 12, 10, 6, 8, 7, 5

2) DFS su G con nuovo ordine.



1: 1

0: 0 → 5 → 4 → 2 → 3

11: 11 → 12 → 9 → 10

6: 6 → 8

7: 7